

$$1a) \ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -kt \quad C_{A0} - C_{A\infty} = 0.1823 - 0.0495 = 0.1328 \text{ g/L}$$

t min	C_A (g/L)	ln	$k = -\ln(c)/t \text{ (min}^{-1}\text{)}$
35	0.1453	-0.326	0.00906
65	0.1216	-0.611	0.00940
100	0.1025	-0.920	0.00920
160	0.075	-1.487	0.00929

$$K = 9.3 \cdot 10^{-3} \text{ min}^{-1}$$

b) C_A @ 120 min $\rightarrow \ln \left(\frac{C_A - 0.049}{0.1328} \right) = -0.0078(128) = 0.092 \text{ g/L}$
 $= 30.5 \text{ gal} \cdot 3.785 = 115.4 \text{ L tank}$

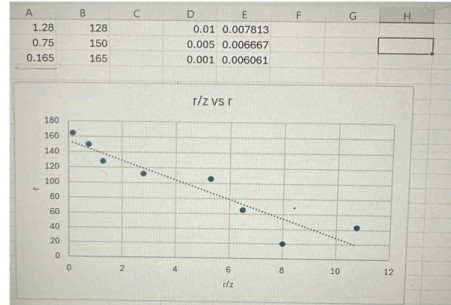
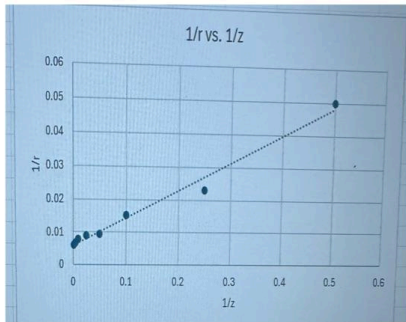
$$\text{Mass A: } 0.092 \cdot 115.4 = 10.6 \text{ g}$$

$$\text{Mass A}_i: 0.1823 \cdot 115.4 = 21 \text{ g}$$

$$21 - 10.6 = 10.4 \text{ g B}$$

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2a) $\frac{z}{r} = \frac{1}{A} z + \frac{B}{A}$ $\uparrow$ z/r vs. z $\frac{1}{r} = \frac{B}{A} \frac{1}{z} + \frac{1}{A}$
 $A = 175$ $B = 9$ $\downarrow$
 plot $1/z$ $1/r$



2b) use Excel solver from $A = 165$ $B = 8.7$ $\times$

5 pts

T	U	V	W
40	128	24.28571	0
100	150	35.78947	0
1000	165	789	0

20125

3) Benzene $\rightarrow$ adipic acid
 0.66 kg waste/kg product

Glucose $\rightarrow$ Adipic acid
 3.77 kg waste/kg product

$$\frac{3.77}{0.66} = 5.7 \quad P_{\text{glucose}} = \frac{1}{5.7} P_{\text{benzene}}$$

Ben. $\rightarrow$ rate.
 0.16 kg/waste kg
 Glu. $\rightarrow$ rate
 5 kg waste/kg

$$\frac{5}{0.16} = 31.25 \quad P_{\text{glucose}} = \frac{1}{31} P_{\text{benzene}}$$

need to be 5.7
 cheaper to
 compete for
 adipic acid

find actual
 prices
 -10 pts

30x cheaper to compete

4) 1970: $1.1 \cdot 10^{12}$ miles
 2000: $2.6 \cdot 10^{12}$ miles
 Emissions: 14 g/m ($1.1 \cdot 10^{12}$ miles)
 1970 $= 15.4 \cdot 10^{12}$ g
 Emissions: 1 g/m ($2.6 \cdot 10^{12}$ miles)
 2000 $= 2.6 \cdot 10^{12}$ g

1970:
 $15.4 \cdot 10^{12}$ g tons $\rightarrow 15.4 \cdot 10^{12} / 907,185 = 1.7 \cdot 10^7$ tons

2000:
 $2.6 \cdot 10^{12}$ g tons $\rightarrow 2.6 \cdot 10^{12} / 907,185 = 2.87 \cdot 10^6$ tons

Percent change:
 $\frac{2.87 \cdot 10^6 - 1.7 \cdot 10^7}{1.7 \cdot 10^7} \cdot 100 = -83.1$

1515

$$5 a) \text{ Masse} = 150.46 = 6900g$$

$$\frac{6900g}{0.40} = 17250g$$

$$\text{Mass W} = 17250g$$

$$\text{Ethanol: } \frac{6900}{0.789} = 8.75L \quad \text{Water: } \frac{10350}{1} = 10.35$$

$$\begin{array}{r} 8.75L \\ + 10.35 \\ \hline 19.1L \end{array} \quad \checkmark$$

$$b) \text{ dens: } 0.93518 g/cm^3$$

$$V = \frac{17250g}{0.93518g/cm^3} = 18.45L$$

$$= \frac{19.1L - 18.45L}{18.45L} \cdot 100$$

$$= 3.5\% \quad \checkmark$$

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